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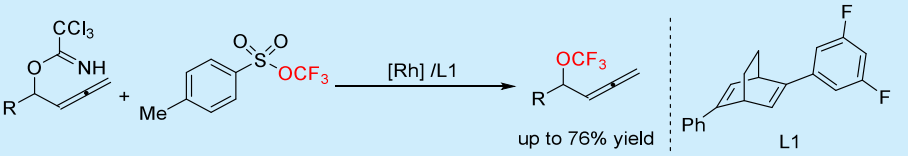
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Rhodium-Catalyzed Trifluoromethoxylation of Allenylic Trichloroacetimidates

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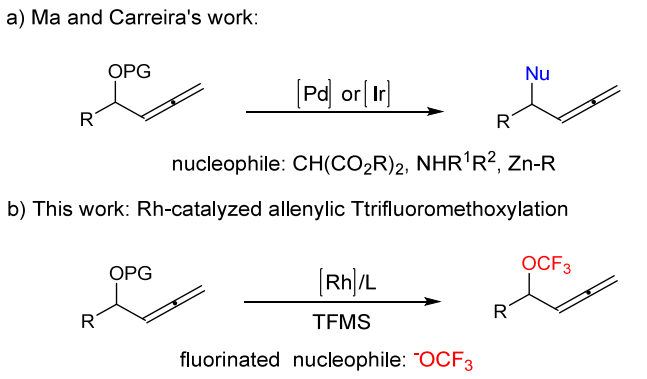
ABSTRACT: The first rhodium-catalyzed trifluoromethoxylation of allenylic trichloroacetimidates has been reported. The products were obtained through a procedure using trifluoromethyl arylsulfonate (TFMS) as the trifluoromethoxy source with a yield up to 76% by identifying the ligand of diene. The reaction proceeds under mild reaction conditions with excellent functional group tolerance and complete regiocontrol.

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Fluorinated compounds are receiving increasing attention because the introduction of a fluorinated functional group into compounds can significantly influence their lipophilicity, which is extremely important in drug molecules, pesticides, and materials.[1] In particular, the OCF_3 group has one of the highest lipophilicity values (π_x = 1.04) compared to those of CF_3 (π_x = 0.88) and SO_2CF_3 (π_x = 0.55). Therefore, the unique peculiarity of trifluoromethoxy intrigues chemists to develop catalytic methods for efficient synthesis of trifluoromethoxylated molecules.[2] However, due to the easy decomposition of trifluoromethoxy into difluorophosgene and fluoride as well as β-fluoride elimination from transition-metal-trifluoromethoxide complexes, introducing this functional group remains a major challenge, especially the site-selective trifluoromethoxylation on a precise molecule.[3]

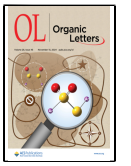
Allenes exist in many natural products and pharmaceuticals.[4] The uniqueness that accumulated double bonds along with high activity make it widely used as synthons for cyclization,[5] hydrogenheteroatom reactions,[6] and multifunctionalization.[7] The versatile reactivity of allenes allows for the selective modification of different positions, enabling the synthesis of highly functionalized compounds.[8] Palladium catalysts were used to catalyze the conversion of allenylic derivatives. The palladium intermediate is attacked via an outer-sphere pathway using soft nucleophiles.[9] When using hard nucleophiles, the 1,3-dienes are obtained through an inner-sphere pathway.[10] Nevertheless, the reactions that structurally maintain allene are much less explored. Ma and co-workers reported palladium catalyzed various conversions of α-centra carbon of allenes.[11] Carreira's group achieved various functionalization of allenylic carbonates using iridium catalysts (Scheme 1a).[12] However, the functionalization of α-centra fluorinated allenes has not yet been reported. Given some of our previous work,[13] we are greatly encouraged to explore the trifluoromethoxy substitution reaction of α-centra secondary carbon in allenes. Herein, we report our efforts to develop the first trifluoromethoxy substitution reaction of allenylic trichloroacetimidates using a Rh catalyst (Scheme 1b).

Scheme 1. Catalytic Strategies to Allenylic Functionalization



Different leaving groups significantly affect the activity of the substrates. We began our studies by choosing the alkyl secondary allenylic ester as the model substrate, which showed high stability that can be recovered completely (See Table S1 in the Supporting Information). Certainly, we attempted to use Pd and Ir for trifluoromethoxylation of allenylic esters but failed (see Table S1 in the Supporting Information).

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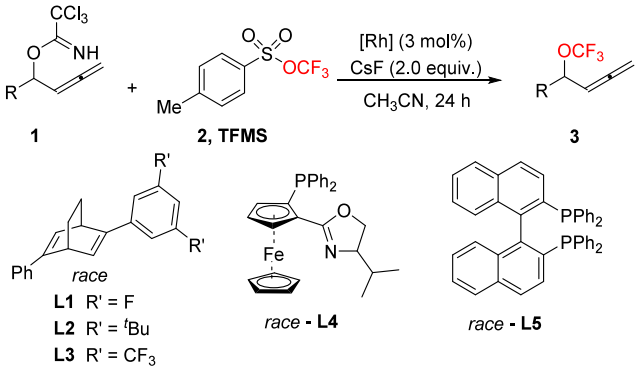
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Trichloroacetimidate is a good leaving group and normally used in the synthesis of glycosides.[14] Moreover, Nguyen and co-workers achieved amination of allylic trichloroacetimidate using rhodium catalyst.[15] Ultimately, by switching the leaving group from Ac and Boc to trichloroacetimidate, a slight yield of 8% was finally obtained (entry 1 in Table 1). Silver salts were added to activate the catalyst by removing the chlorine.[16] As expected, a yield of 36% was furnished with 0.3 equiv of AgF (entry 2). Lowering the reaction temperature to −40 °C resulted in a slight increase in the yield (entry 3). After screening the equivalent of AgF, a yield of 49% was achieved using 0.5 equiv of AgF as an additive (entry 5). Other silver salts can lead to a slight decrease in yield (entries 7 to 9). The yield ascends to 58% with $TBABF_4$ as a phase transfer catalyst (entry 11). Replacing the diene ligand of rhodium from cyclooctadiene to **L1** delivered a slight increase in yield with 62% similar to **L3** (entry 13 and 15), while the use of electron-rich diene **L2** resulted in a decrease in the yield (entry 14). The other ligands **L4** and **L5** afforded almost no product. It is worth noting that the reaction proceeded with completely regiocontrol.

With the optimized condition in hand (entry 13 in Table 1), we began to examine the scope of allenylic trichloroacetimidates (Scheme 2). Generally, both electron-donating and electron-withdrawing groups including OMe, CF_3, and NO_2 were well tolerated, providing the corresponding products in 60−62% yields (**3a**−**3c**). Substrates that contain halogen can be applied to this system, which allowed further decoration (53−65%, for **3d** and **3e**). However, the sulfonamide substrate provided the product with a higher yield (70%, **3f**), probably due to the coordination of sulfone. Substrates bearing

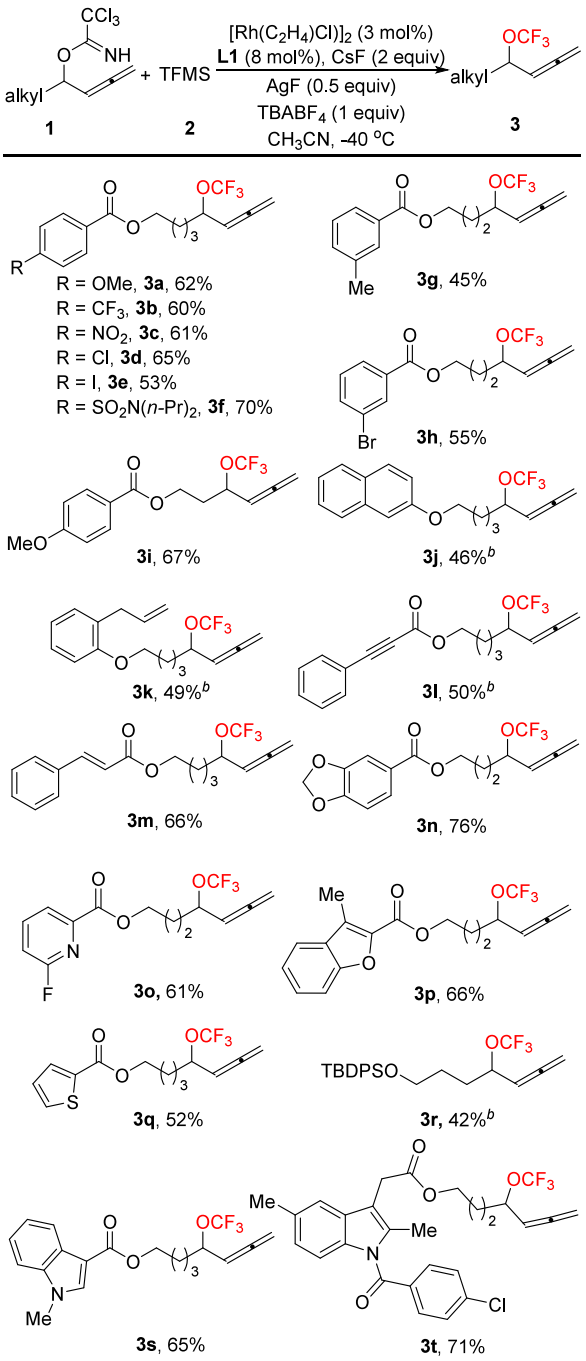
Table 1. Optimization of Reaction Conditions[a]



entry	[Rh]	additive	PTC	yield/%
1[b]	$[Rh(cod)Cl]_2$			8
2[b]	$[Rh(cod)Cl]_2$	AgF (0.3 equiv)		36
3	$[Rh(cod)Cl]_2$	AgF (0.3 equiv)		40
4[c]	$[Rh(cod)Cl]_2$	AgF (0.3 equiv)		45
5[c]	$[Rh(cod)Cl]_2$	AgF (0.5 equiv)		49
6[c]	$[Rh(cod)Cl]_2$	AgF (1.0 equiv)		48
7[c,d]	$[Rh(cod)Cl]_2$	AgOTf		41
8[c,d]	$[Rh(cod)Cl]_2$	$AgBF_4$		42
9[c,d]	$[Rh(cod)Cl]_2$	$AgClO_4$		48
10[c,d]	$[Rh(cod)Cl]_2$	AgF	$TBAPF_6$	50
11[c,d]	$[Rh(cod)Cl]_2$	AgF	$TBABF_4$	58
12[c,d]	$[Rh(cod)Cl]_2$	AgF	$(Ph)_4PBF_4$	42
13[c,d]	$[Rh(C_2H_2)Cl]_2$+**L1**	AgF	$TBABF_4$	62
14[c,d]	$[Rh(C_2H_2)Cl]_2$+**L2**	AgF	$TBABF_4$	24
15[c,d]	$[Rh(C_2H_2)Cl]_2$+**L3**	AgF	$TBABF_4$	60
16[c,d]	$[Rh(C_2H_2)Cl]_2$+**L4**	AgF	$TBABF_4$	6
17[c,d]	$[Rh(C_2H_2)Cl]_2$+**L5**	AgF	$TBABF_4$	2

[a]Reaction conditions: **1** (0.3 mmol), TFMS (3 equiv), $[Rh(C_2H_4)Cl]_2$ (3 mol %), L (8 mol %), CsF (2.0 equiv), CH_3CN (3 mL), −40 °C, 24 h. Yields were determined by ^{19}FNMR spectroscopy with benzotrifluoride as a standard. R = *p*-OMe-C_6H_4-CO_2-$(CH_2)_4$. [b]0 °C. [c]TFMS (4 equiv). [d][Ag] (0.5 equiv).

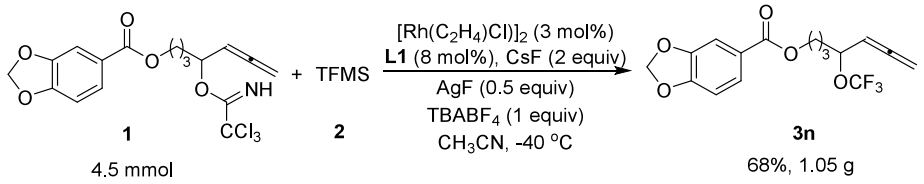
Scheme 2. Substrate Scope of Trifluoromethoxylation[a]



[a]Reaction conditions: **1** (0.3 mmol), TFMS (4 equiv), $[Rh(C_2H_4)Cl]_2$ (3 mol %), L1 (8 mol %), CsF (2.0 equiv), CH_3CN (3 mL), −40 °C, 24 h. Yields of isolated products are given, unless otherwise noted. [b]$[Rh(cod)Cl]_2$ (3 mol %), **1** (0.3 mmol), CsF (2.0 equiv), **2** (4.0 equiv), CH_3CN (3 mL), −40 °C, 24 h.

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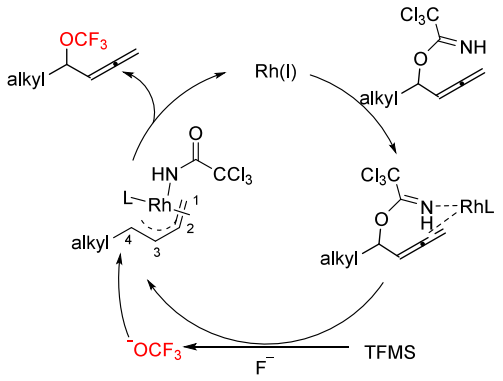
Scheme 3. Gram-Scale Synthesis



functional groups near the allenylic site afforded medium yields (45−55%, **3g** and **3h**). In addition, compared with **3a**, the yield of **3i** increased to 67%, which may be due to the greater electron-withdrawing effect of the ester group. Besides the ester group, ether can also be a connecting motif, which is helpful to introduce more different groups into the substrates and greatly enrich the diversity of products (46−49%, **3j** and **3k**). This method also worked well with the allylic group, and the products were isolated with moderate yield (**3k** and **3m**). Notably, the alkyne can also be compatible with this catalytic system (**3l**) in 50% yield. To our delight, heterocyclic substrates can also be tolerated, affording the target products in 52−76% yields (**3n**−**3q**). Since there are a lot of fluoride anions in the system, silicon-containing groups probably will not survive. Surprisingly, the steric hindered $^{t}BuPh_2Si$ remained and the target product was provided with a yield of 42% (**3r**). Finally, we evaluated the potential of this catalytic trifluoromethoxylation method for indole and its derivatives, which delivered the corresponding product in 65% and 71% yield, respectively. A gram-scale reaction was conducted, and **3n** was prepared in 68% yield (Scheme 3).

Based on the related literature[15,17] and the recent work[18] by our group, we suggest the mechanism shown in Scheme 4. It is

Scheme 4. Proposed Mechanism



envisioned that a plausible mechanism would arise from coordination of the nitrogen and the double bond of allenylic trichloroacetimidate to the rhodium catalyst to form the activated rhodium allenylic complex. Reductive elimination of trifluoromethoxy should be a tough process as the most favored β-fluoride elimination would take place from transition-metal-trifluoromethoxide complexes. We consider that the steric hindrance created by the ligands around the Rh center would preclude the reaction of an external nucleophile at C2; therefore, the favored reaction is observed to occur at C4.[12a] This reaction proceeds with excellent regioselectivity, which may be attributed to the outer-sphere mechanism of attacking the C4.

In conclusion, we have presented the first example of rhodium-catalyzed trifluoromethoxylation of allenylic trichloroacetimidates using trifluoromethyl arylsulfonate (TFMS) as an OCF_3 source. The combination of rhodium with diene ligand exhibited excellent catalytic performance, providing a range of allenylic trifluoromethoxy compounds with up to 76% yield and complete regiocontrol, which are difficult to access by other methods. Due to the easy decomposition of the OCF_3 group, carefully selecting the appropriate leaving group, ligand structure, reaction temperature, solvent, and base was very critical, as all the factors jointly guaranteed the smooth transformation. This discovery also expanded the scope of the application of trifluoromethoxy reagents.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.4c03676.

Experimental materials and procedures, characterizations, and copies of NMR spectra of compounds (PDF)

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Notes

The authors declare no competing financial interest.

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